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Business Requirements Document: AI-Native Analytics Platform Initiative

Executive Overview

This document outlines the business case for transforming the company's analytical infrastructure from a legacy PostgreSQL-based system to a modern, AI-Native analytical platform. The initiative addresses critical business pain points that are directly impacting revenue, operational efficiency, and competitive positioning in the market.

The proposed solution will reduce operational costs, accelerate time-to-insight, and enable data-driven decision-making at scale. This transformation is not merely a technical upgrade; it is a strategic business initiative that positions the company for sustainable growth in an increasingly data-driven marketplace.

1. Business Context & Market Position

1.1 Current Market Landscape

The e-commerce industry is experiencing unprecedented growth, with companies competing on speed, personalization, and operational efficiency. Data has become the primary competitive advantage. Companies that can analyze customer behavior, optimize inventory, and respond to market trends in real-time are winning market share.

Our company currently operates in a highly competitive segment where margins are thin and customer expectations are high. To maintain market position and grow revenue, we must be able to:

- **Respond to market trends within hours, not days**
- **Personalize customer experiences based on real-time behavioral data**
- **Optimize pricing and inventory dynamically**

- **Detect and respond to operational anomalies before they impact customers**

1.2 The Competitive Disadvantage

Our current analytics infrastructure is built on PostgreSQL, a transactional database never designed for analytical workloads. This architectural mismatch is creating a competitive disadvantage:

- **Slower insights:** Analytics queries take minutes to hours to execute, delaying decision-making
- **Operational friction:** Data engineers spend 60-70% of their time maintaining the broken system rather than building new capabilities
- **Limited scalability:** The system struggles during peak traffic periods, precisely when analytics are most valuable
- **Reduced agility:** New analytical requirements take weeks to implement due to infrastructure constraints

Competitors using modern analytics stacks (Snowflake, BigQuery, Databricks) are making faster decisions and launching new features more quickly. We are falling behind.

2. The Problem Statement: Business Impact

2.1 Revenue Impact

The current analytics infrastructure is directly impacting revenue in three ways:

Lost Revenue from Slow Insights

During peak sales events (Black Friday, holiday season), our ability to analyze customer behavior and adjust pricing/promotions in real-time is severely limited. Competitors using modern analytics platforms can adjust prices every 15 minutes based on demand signals. We adjust prices every 24 hours due to infrastructure constraints. This translates to an estimated **\$2-5M in lost revenue annually** from suboptimal pricing decisions.

Missed Cross-Sell Opportunities

Our recommendation engine relies on batch-processed analytics that runs once per day. Modern competitors use real-time analytics to personalize recommendations on every page view. This results in approximately **15-20% lower average order value** compared to competitors, representing **\$3-8M in lost annual revenue**.

Customer Churn from Poor Experience

When our analytics infrastructure fails (which happens 2-3 times per month during peak traffic), our dashboards go down and our customer support team loses visibility into

customer issues. This leads to delayed response times and increased churn. We estimate this costs us **\$1-3M annually** in lost customer lifetime value.

Total Estimated Annual Revenue Impact: \$6-16M

2.2 Operational Cost Impact

Data Engineering Productivity Loss

Our data engineering team (8 engineers) spends approximately 60-70% of their time maintaining the broken PostgreSQL analytics system:

- Debugging failed queries
- Manually reconciling data inconsistencies
- Writing post-mortem reports
- Patching schema issues

This means only 30-40% of engineering capacity is available for building new analytical capabilities. At an average cost of \$150K per engineer per year, this represents **\$360-480K in annual productivity loss** (8 engineers × \$150K × 50% of time spent on maintenance).

Incident Response Costs

When the analytics system fails, it triggers a cascade of manual interventions:

- On-call engineers must debug the issue (4-8 hours per incident)
- Business teams must make decisions without data
- Customer support must handle angry customers
- Post-incident reviews and documentation (another 4-8 hours)

At 2-3 incidents per month, this costs approximately **\$80-120K annually** in unplanned labor.

Infrastructure Costs

The current system requires:

- A dedicated PostgreSQL instance sized for both transactional and analytical workloads (overprovisioned for analytics)
- Expensive storage for historical data
- Manual backup and recovery processes
- Estimated annual cost: **\$150-200K**

Total Estimated Annual Operational Cost Impact: \$590-800K

2.3 Strategic Impact

Beyond direct financial impact, the current infrastructure is limiting strategic initiatives:

- **Real-Time Personalization:** Cannot be implemented due to analytics latency
- **Predictive Analytics:** Cannot be scaled due to infrastructure constraints
- **AI-Driven Optimization:** Cannot be deployed due to lack of real-time data access
- **Data Democratization:** Cannot enable business teams to self-serve analytics due to system fragility

These limitations are preventing the company from competing on innovation and are limiting our ability to attract top talent who want to work with modern technology stacks.

3. The Business Solution: AI-Native Analytics Platform

3.1 What We're Building

The proposed solution is a **modern, purpose-built analytical platform** that separates transactional and analytical concerns entirely. This platform will:

- **Extract data** from the transactional PostgreSQL database incrementally
- **Transform data** using industry best practices (dbt + Medallion architecture)
- **Serve data** through a high-performance analytical engine (DuckDB)
- **Expose data** via natural language interfaces (MCP + Claude Desktop)
- **Monitor autonomously** using AI agents (Sentinel Engine)

This is not an incremental improvement; it is a fundamental architectural transformation.

3.2 Business Benefits

Faster Insights (Revenue Impact)

- Analytical queries will execute in **seconds instead of minutes/hours**
- Real-time pricing optimization will be possible, enabling **\$2-5M in additional annual revenue**
- Real-time personalization will increase average order value by **15-20%, adding \$3-8M in annual revenue**
- Estimated total revenue uplift: **\$5-13M annually**

Reduced Operational Costs (Cost Impact)

- Data engineers will spend only **20-30% of time on maintenance** (vs. 60-70% today)
- This frees up **\$240-320K in annual engineering capacity** for building new capabilities

- Incident response time will drop from **4-8 hours to 15-30 minutes** due to autonomous monitoring
- This saves **\$60-100K annually** in unplanned labor
- Infrastructure costs will drop to **\$50-80K annually** (serverless, pay-per-query model)
- Estimated total cost reduction: **\$350-500K annually**

Improved Reliability (Risk Mitigation)

- The Sentinel Engine will detect and remediate issues **before they impact customers**
- System availability will improve from **98% to 99.5%+**
- This reduces customer churn and improves brand reputation
- Estimated value: **\$500K-1M annually** in avoided churn

Strategic Enablement (Future Value)

- Real-time personalization becomes possible
- Predictive analytics can be scaled
- AI-driven optimization can be deployed
- Business teams can access data via natural language
- Estimated future value: **\$10-20M+ annually** (3-5 year horizon)

Total Estimated Annual Business Impact: \$5.85-14.5M in Year 1, growing to \$15-33.5M by Year 3

4. Investment Required

4.1 Development Costs

Engineering Time

- Lead Data Engineer (1 FTE, 4 months): \$50K
- Agentic Engineer (1 FTE, 4 months): \$50K
- Backend Engineer (0.5 FTE, 4 months): \$25K
- **Total Engineering: \$125K**

External Services & Tools

- Dagster Cloud (3-month trial, then production): \$5K
- MotherDuck (initial setup & optimization): \$3K
- LLM API costs (GPT-4o for Sentinel Engine): \$2K

- **Total Services: \$10K**

Total Development Investment: \$135K

4.2 Ongoing Operational Costs (Annual)

Infrastructure

- MotherDuck (compute-based pricing): \$30-50K/year
- Dagster Cloud (serverless tier): \$20-30K/year
- FastAPI hosting (Cloud Run/Lambda): \$5-10K/year
- **Total Infrastructure: \$55-90K/year**

LLM API Costs

- Sentinel Engine (GPT-4o for Manager, GPT-4o-mini for specialists): \$40-60K/year
- **Total LLM Costs: \$40-60K/year**

Maintenance & Support

- 0.5 FTE data engineer for ongoing maintenance: \$75K/year
- **Total Maintenance: \$75K/year**

Total Annual Operational Cost: \$170-225K/year

5. Return on Investment (ROI) Analysis

5.1 Conservative Scenario (Year 1)

Metric	Value
Revenue Uplift	\$5M
Cost Reduction	\$350K
Total Benefit	\$5.35M
Development Investment	\$135K
Operational Cost (Year 1)	\$170K
Total Year 1 Cost	\$305K
Net Year 1 Benefit	\$5.045M
ROI	1,653%

Payback Period	22 days
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5.2 Moderate Scenario (Year 1)

Metric	Value
Revenue Uplift	\$9M
Cost Reduction	\$425K
Total Benefit	\$9.425M
Development Investment	\$135K
Operational Cost (Year 1)	\$200K
Total Year 1 Cost	\$335K
Net Year 1 Benefit	\$9.09M
ROI	2,713%
Payback Period	13 days

5.3 Aggressive Scenario (Year 1)

Metric	Value
Revenue Uplift	\$13M
Cost Reduction	\$500K
Total Benefit	\$13.5M
Development Investment	\$135K
Operational Cost (Year 1)	\$225K
Total Year 1 Cost	\$360K
Net Year 1 Benefit	\$13.14M
ROI	3,650%
Payback Period	10 days

5.4 Multi-Year ROI (Moderate Scenario)

Year	Revenue Uplift	Cost Reduction	Total Benefit	Operational Cost	Net Benefit	Cumulative
Year 1	\$9M	\$425K	\$9.425M	\$200K	\$9.225M	\$9.225M
Year 2	\$15M	\$500K	\$15.5M	\$220K	\$15.28M	\$24.505M
Year 3	\$20M	\$550K	\$20.55M	\$240K	\$20.31M	\$44.815M
3-Year Total	\$44M	\$1.475M	\$45.475M	\$660K	\$44.815M	—

6. Risk Analysis & Mitigation

6.1 Technical Risks

Risk: Implementation delays

- *Probability:* Medium
- *Impact:* High (delays revenue realization)
- *Mitigation:* Use AgentSpec-driven development to accelerate implementation; allocate experienced engineers; establish clear milestones and checkpoints

Risk: Data quality issues during migration

- *Probability:* Medium
- *Impact:* Medium (requires remediation but doesn't block deployment)
- *Mitigation:* Build comprehensive data validation tests; use synthetic data generator to stress-test before production; implement gradual rollout

Risk: Performance issues at scale

- *Probability:* Low
- *Impact:* High (impacts revenue realization)
- *Mitigation:* Conduct thorough load testing; use DuckDB's proven scalability; implement monitoring and alerting

6.2 Business Risks

Risk: Adoption by business teams

- *Probability:* Low
- *Impact:* Medium (limits revenue uplift realization)
- *Mitigation:* Provide training and documentation; start with high-impact use cases; build MCP interface for ease of use

Risk: Competitive response

- *Probability:* High
- *Impact:* Low (we're catching up, not ahead)
- *Mitigation:* Execute quickly; focus on operational excellence; build sustainable competitive advantage through data culture

6.3 Organizational Risks

Risk: Resistance from current analytics team

- *Probability:* Medium
- *Impact:* Medium (slows adoption)
- *Mitigation:* Involve team in design; emphasize that this frees them from maintenance to build new capabilities; provide training and support

Risk: Knowledge gaps in AI-Native engineering

- *Probability:* Medium
- *Impact:* Medium (requires external expertise)
- *Mitigation:* Hire or contract experienced agentic engineers; invest in team training; use AgentSpec framework to codify best practices

7. Success Metrics & KPIs

7.1 Business Metrics

KPI	Current State	Target (Year 1)	Target (Year 2)
Average Order Value	\$75	\$86 (+15%)	\$90 (+20%)
Pricing Optimization Frequency	1x/day	15-min intervals	Real-time
Analytics Query Latency	5-30 min	<5 sec	<1 sec

Data-Driven Decision Velocity	1-2 per week	10-15 per week	50+ per week
Customer Churn Rate	5%	4%	3%
System Availability	98%	99.5%	99.9%

7.2 Operational Metrics

KPI	Current State	Target (Year 1)
Data Engineer Maintenance Time	60-70%	20-30%
Mean Time to Resolution (MTTR)	4-8 hours	15-30 minutes
Analytics Incidents per Month	2-3	<0.5
Cost per Analytical Query	\$0.50	\$0.05
Data Freshness	24 hours	Real-time

7.3 Strategic Metrics

KPI	Target (Year 1)	Target (Year 2)
Real-Time Personalization Campaigns	0	10+
Predictive Models in Production	0	5+
Business Users with Self-Service Analytics	0	30+
New Analytical Capabilities Launched	2-3	15+

8. Implementation Timeline

Phase 1: Validation (Weeks 1-2)

- Build synthetic data generator
- Validate platform architecture with realistic failure scenarios
- Stress test the entire system
- Obtain stakeholder approval to proceed

Phase 2: Development (Weeks 3-8)

- Build Analytical Platform (Dagster + dbt + DuckDB)
- Build MCP server for natural language querying
- Build Sentinel Engine (CrewAI multi-agent system)
- Conduct comprehensive testing

Phase 3: Pilot (Weeks 9-12)

- Deploy to pilot environment
- Run parallel with existing system
- Validate business metrics
- Train pilot users

Phase 4: Production Rollout (Weeks 13-16)

- Gradual migration of analytics workloads
- Monitor performance and reliability
- Optimize based on real-world usage
- Decommission legacy system

Phase 5: Optimization & Scale (Weeks 17+)

- Optimize for cost and performance
 - Enable advanced use cases (real-time personalization, predictive analytics)
 - Scale to additional data sources
 - Build organizational analytics capability
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9. Stakeholder Impact

9.1 Executive Leadership

- **Benefit:** Improved profitability (\$5-13M annual revenue uplift, \$350-500K cost reduction)
- **Benefit:** Reduced operational risk (autonomous monitoring, faster incident response)
- **Benefit:** Strategic enablement (real-time personalization, predictive analytics)
- **Commitment:** Allocate budget (\$135K development, \$170-225K annual operations); approve timeline

9.2 Data Engineering Team

- **Benefit:** Shift from maintenance to innovation (60-70% of time freed up)
- **Benefit:** Work with modern technology stack (Dagster, dbt, DuckDB, CrewAI)
- **Benefit:** Reduced on-call burden (autonomous monitoring)
- **Challenge:** Learning curve for new tools and AI-Native paradigm
- **Commitment:** Participate in design and implementation; invest in learning

9.3 Analytics & Business Intelligence Team

- **Benefit:** Faster insights (seconds instead of minutes/hours)
- **Benefit:** Self-service analytics via natural language (Claude Desktop)
- **Benefit:** More reliable system (fewer incidents)
- **Challenge:** Learning new tools and interfaces
- **Commitment:** Participate in pilot; provide feedback; adopt new tools

9.4 Product & Marketing Teams

- **Benefit:** Real-time insights for decision-making
- **Benefit:** Ability to test and optimize campaigns in real-time
- **Benefit:** Better customer understanding (real-time personalization)
- **Challenge:** Requires behavior change (moving from weekly reports to real-time queries)
- **Commitment:** Engage with analytics team; adopt new tools; drive adoption of insights

9.5 Finance & Accounting

- **Benefit:** Better cost visibility (pay-per-query model)
- **Benefit:** Reduced unplanned incident costs
- **Benefit:** Improved profitability

- **Challenge:** New cost structure (shift from CapEx to OpEx)
 - **Commitment:** Approve budget; adjust financial models
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10. Decision Framework

10.1 Go/No-Go Criteria

Go Decision If:

-  Development team confirms technical feasibility
-  ROI analysis shows positive payback within 12 months
-  Executive leadership approves budget and timeline
-  Key stakeholders (Data, Analytics, Product) commit to participation
-  No critical blockers identified in risk assessment

No-Go Decision If:

-  Technical feasibility cannot be confirmed
-  ROI is negative or payback exceeds 24 months
-  Executive leadership withholds budget or timeline approval
-  Key stakeholders refuse to participate
-  Critical technical or organizational risks cannot be mitigated

10.2 Approval Sign-Off

Role	Name	Signature	Date
Chief Technology Officer	—	—	—
Chief Financial Officer	—	—	—
VP of Product	—	—	—
VP of Data & Analytics	—	—	—
Head of Engineering	—	—	—

11. Appendix: Supporting Data

A. Industry Benchmarks

Modern analytics platforms (Snowflake, BigQuery, Databricks) enable:

- Query latency: <5 seconds for 99th percentile queries
- System availability: 99.9%+
- Cost per query: \$0.01-0.10 (vs. our current \$0.50)
- Data freshness: Real-time to minutes
- Time to new analytical capability: 1-2 weeks (vs. our current 3-4 weeks)

B. Competitive Analysis

Competitor	Analytics Platform	Pricing Optimization	Personalization	Time-to-Insight
Competitor A	BigQuery	Real-time	Real-time	<1 min
Competitor B	Snowflake	15-min intervals	Real-time	<2 min
Competitor C	Databricks	Real-time	Real-time	<1 min
Our Company (Current)	PostgreSQL	1x/day	Batch	5-30 min
Our Company (Proposed)	DuckDB + Dagster	Real-time	Real-time	<5 sec

C. Customer Impact Examples

Example 1: Real-Time Pricing Optimization

- Competitor adjusts prices every 15 minutes based on demand
- We adjust prices once per day
- During a 10-hour peak sales window, competitor captures 15-20% more revenue through dynamic pricing
- Annual impact: \$2-5M

Example 2: Real-Time Personalization

- Competitor shows personalized recommendations on every page view

- We show batch-processed recommendations updated once per day
- Competitor achieves 20% higher average order value
- Annual impact: \$3-8M

Example 3: Operational Resilience

- Competitor detects and resolves issues in 15-30 minutes
 - We detect and resolve issues in 4-8 hours
 - During an incident, competitor loses 1 hour of revenue; we lose 4-8 hours
 - Annual impact (2-3 incidents/month): \$1-3M
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12. Conclusion

The proposed AI-Native Analytics Platform is not just a technical upgrade; it is a strategic business initiative that will:

- **Increase revenue by \$5-13M annually** through faster insights and real-time optimization
- **Reduce costs by \$350-500K annually** through operational efficiency
- **Improve reliability and reduce risk** through autonomous monitoring
- **Enable future innovation** through modern data architecture
- **Improve competitive positioning** in a data-driven market

With an ROI of 1,600-3,650% and a payback period of 10-22 days, this initiative represents one of the highest-return investments the company can make.

Recommendation: Proceed with implementation immediately.

This document is intended for internal use by executive leadership, key stakeholders, and the implementation team. It should be treated as confidential business information.